

Power Cost Equalization - Data Cleaning

Methodology Notes

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Power Cost Equalization Data Cleaning Process

The Power Cost Equalization (PCE) program was created in 1984 when the Alaska Legislature enacted the Alaska Statutes 44.83.162-165. The program provides economic assistance to many rural Alaska communities by reducing electricity rates charged to eligible rural residents. PCE covers 95% of a utility's cost between a floor or base rate (equal to the weighted average rate for Anchorage, Fairbanks and Juneau) and a ceiling (set by the Legislature) for a defined level of consumption (currently, 500 kilowatt-hours for residential customers and 70 kilowatt-hours per month multiplied by the community's population for public facilities).

The program is administered by the Regulatory Commission of Alaska (RCA), which sets the PCE rate, and the Alaska Energy Authority (AEA), which distributes payments to the eligible and participating utilities. As part of the program, the utilities report information about their operation to RCA and AEA.

Participants of the Power Cost Equalization (PCE) program report to the Regulatory Commission of Alaska (RCA) for fuel cost adjustments to their rates. The RCA has authority to maintain accounts and records of public utilities that fall under its jurisdiction, under Alaska Statute 42.05.451. This responsibility allows the Regulatory Commission of Alaska to obtain information from regulated utilities. Additionally, all utilities that serve ten or more customers must obtain an operating certificate, which describes the authorized service area and scope of operations of the utility. The RCA will issue a certificate when it finds the utility to be fit, willing, and able to provide the service. The RCA maintains a list of both regulated and unregulated certified utilities. Utilities report annually to the RCA, but file a PCE Utility Monthly Report with AEA. However, though utilities are required to submit the same information to both agencies, there is no formal reconciliation of data between RCA and AEA.

AEA stores the data they receive from utilities in the PCE Utility Monthly Report in their financial data system called NAVISION. This system uses an accounting framework where transactions are recorded. The Institute of Social and Economic Research (ISER) receives PCE data files from the Alaska Energy Authority. ISER has processed these files to present the data in a user-friendly format and to eliminate as many erroneous data points as possible. However, the guiding principle during this process is to maintain the original data as intact as possible. A detailed description of methodology used to adjust the PCE data to a more user-friendly format is presented below.

- a) Variables that have no informative value to users beyond the administrative/accounting staff at the Alaska Energy Authority are not included. (Please refer to the Variable Dictionary on page 3).
- b) Duplicate observations are deleted.
- c) Variable names are changed to be more descriptive and to conform to the variable naming convention in the Alaska Energy Data Gateway and the Alaska Energy Statistics publications.
- d) Community names are added and standardized to reflect those used by the U.S. Census and to include multiple communities when applicable.
- e) Utilities may report to AEA on a monthly basis. However, utilities sometimes do not participate in the program all year or may fail to submit data for a particular month. When a utility does not submit a monthly report, no record is created in the NAVISION data set. Hence, some

utilities/communities have less than twelve records per year. We add observations so that all utilities/communities have twelve records per year by creating observations with missing data points where appropriate (also called balancing the panel).

- f) Variables are added to the data set such as month, year, effective rate, intertie and others. Please refer to the PCE Variable Dictionary for details.
- g) Illogical and or erroneous data points are identified and deleted.
 - The PCE data set does not differentiate between missing and zero values, so we attempt to make that distinction when possible.
 - Variable values are changed to missing values in the following cases:
 - Residential rates, if effective rates are below 8 cents/kWh.
 - Non fuel expenses, if equal or less than zero.
 - Powerhouse consumption, if equal or less than zero.
 - Peak consumption, if equal or less than zero.
 - Fuel price, if equal or less than zero; and less than a floor (less than 50 cents per gallon) and more than a ceiling (\$10 per gallon)
 - Diesel kWh generated, if equal or less than zero, unless it refers to an intertied community in which the value zero is preserved.
 - Residential kWh sold, if less than or equal to zero.
 - Commercial, Community, Government and Unbilled kWh sold, if less than zero.
 - For fuel use, fuel cost and diesel kWh generated if values are not within two standard deviations and have efficiency values of less than 5 kWh/gal or more than 20 kWh/gal
 - The Alaska Village Electric Cooperative and the Alaska Power Company report non-fuel expenses annually and the annual value is recorded in the month of June. The annual amount is distributed equally over twelve months of the year.

PCE Variable Dictionary

NAVISION Variable Name	AEDG Variable Name	Variable Description
Buy-from Vendor No.	utility_acronym	Acronym used in the Power Cost Equalization (NAVISION) database to identify the utility that received payment for an invoice. Variable name in AEA's PCE NAVISION system: Buy-from Vendor No.
Project Code	pce_id	Unique identifier in the PCE database (called Project Code in the Alaska Energy Authority's NAVISION system). It represents a community, or set of intertied communities, if one report for reimbursement is filed on behalf of one or multiple communities. For example, INN Electric Cooperative reports and receives a single payment for the communities of Iliamna, Newhalen and Nondalton. So, this identifier represents all three communities in this dataset. Variable name in AEA's PCE NAVISION system: Project Code.
Posting Description	community_name	Name of community (ies) as recorded in the U.S. Census that reported that year to the Alaska Energy Authority to receive Power Cost Equalization payment. Variable name in AEA's PCE NAVISION system: Posting Description.
	Intertie	Variable added by ISER. It indicates if a community has an intertie to one or more communities in which all communities in the intertie depend on a shared electricity system.
	Year	Variable added by ISER.
	Month	Variable added by ISER.
	Season	Variable added by ISER. Value 'Summer' assigned to months from April to September; value 'Winter' assigned to months October to March.
Residential Rate	residential_rate	Rate, in dollars per kilowatt-hour, charged to residential customers for the first 500 kWh as reported by the utility and prior to adjusting to reflect PCE assistance. Variable name in AEA's PCE NAVISION system: Residential Rate.

Actual Rate	pce_rate	Power Cost Equalization rate (\$/kWh) approved by the Regulatory Commission of Alaska that the Alaska Energy Authority uses to calculate the total PCE payment. This amount is subtracted from the residential rate for the first 500 kilowatt-hours a residential customer consumes in a month. Variable name in AEA's PCE NAVISION system: Actual Rate.
Pro Rata Rate	pro_rata_rate	Power Cost Equalization rate (\$/kWh) adjusted when the program is not fully funded by the Alaska State Legislature. When full funding is available the PCE pro-rata rate equals the PCE rate. Variable name in AEA's PCE NAVISION system: Pro Rata Rate.
	effective_rate	Variable added by ISER. Calculated variable by subtracting the PCE pro-rata rate from the residential rate. This is the rate a residential customer pays for the first 500 kilowatt-hours consumed in a month.
PCE Eligible Residential kWh	pce_eligible_residential_kwh	Total monthly kilowatt-hours consumed by residential customers eligible to receive PCE payment, as determined by the Alaska Energy Authority. Each customer is eligible to receive PCE payment for the first 500 kilowatt-hours consumed in a month. Variable name in AEA's PCE NAVISION system: PCE Eligible Residential kWh.
PCE Eligible Commercial kWh	pce_eligible_commercial_kwh	Total monthly kilowatt-hours consumed by commercial customers eligible to receive PCE payment. Since FY1999 commercial customers are not eligible to receive PCE payments, hence reported dollars for all communities should be zero. Variable name in AEA's PCE NAVISION system: PCE Eligible Commercial kWh.
PCE Eligible Community kWh	pce_eligible_community_kwh	Total monthly kilowatt-hours consumed at community facilities eligible to receive PCE payment, as determined by the Alaska Energy Authority. The eligible kilowatt-hour limit is calculated by multiplying the number of residents in a community by 70 kilowatt-hours. Community facilities is defined as water and sewer facilities, charitable educational facilities, public lighting, or community buildings whose operations are not paid by the state, federal government or private commercial party. Variable name in AEA's PCE NAVISION system: PCE Eligible Community kWh.

PCE Eligible kWh Total	pce_eligible_total_kwh	Total monthly kilowatt-hours eligible to receive PCE payments. The sum of pce_eligible_residential_kwh, pce_eligible_commercial_kwh and pce_eligible_community_kwh. Variable name in AEA's PCE NAVISION system: PCE Eligible kWh Total.
Amount	disbursement	Total monthly PCE amount paid to the utility for PCE eligible kilowatt-hours sold; includes any payment adjustments (positive or negative) recorded for that month; as determined by the Alaska Energy Authority. Variable name in AEA's PCE NAVISION system: Amount.
Most Recent Fuel Purch. Price	fuel_price	Most recent fuel purchase price of diesel/fuel oil used for electricity generation reported by the utility to the Alaska Energy Authority. Variable name in AEA's PCE NAVISION system: Most Recent Fuel Purch. Price.
Fuel Used (Gallons)	fuel_used_gal	Total gallons used for electricity generation by the utility as reported to the Alaska Energy Authority. Variable name in AEA's PCE NAVISION system: Fuel Used (Gallons).
Fuel Cost	fuel_cost	Total fuel costs (in dollars) paid by the utility for purchased fuel as reported to the Alaska Energy Authority. Variable name in AEA's PCE NAVISION system: Fuel Cost.
Non-Fuel Expenses	nonfuel_expenses	Total non-fuel expenses as reported by the utility to the Alaska Energy Authority; in dollars. Non fuel costs include: personnel expenses (e.g. wages, employer FICA-FUTA-SUI, casual labor, workers compensation insurance), operating costs (e.g. oil, filters, repairs and maintenance, tools, equipment rental), general and administrative costs (professional services (accountants, lawyers, engineers, etc.), insurance, office supplies, postage, office rent, travel, training, bad debt expense, RCA fees), interest expense (from a year end bank statement or loan amortization), depreciation (of assets such as engines, generators, fuel storage tanks, utility poles, lines, transformers, meters, computers, billing software, office furniture, vehicles, buildings). Variable name in AEA's PCE NAVISION system: Non-Fuel Expenses.

Diesel kWh Generated	diesel_kwh_generated	Total kilowatt-hours produced from all generators burning diesel/fuel oil, as reported by the utility to the Alaska Energy Authority (including kwh used in the power plant). Variable name in AEA's PCE NAVISION system: Purchased From.
Hydro kWh Generated	hydro_kwh_generated	Total monthly kilowatt-hours generated from hydroelectric resources as reported by the utility to the Alaska Energy Authority. Variable name in AEA's PCE NAVISION system: Hydro kWh Generated.
Other 1 kWh Type	other_1_kwh_type	Energy source description of kilowatt-hours produced from sources other than diesel/fuel oil or hydroelectric power. Description listed here corresponds to kilowatt-hours listed under 'other_1_kwh_generated'. Variable name in AEA's PCE NAVISION system: Other 1 kWh Type.
Other 1 kWh Generated	other_1_kwh_generated	Total kilowatt-hours produced from fuels different than diesel/fuel oil or hydroelectric power. For example, natural gas or wind. Please refer to 'other_1_kwh_type' for the corresponding fuel type description. Variable name in AEA's PCE NAVISION system: Other 1 kWh Generated.
Other 2 kWh Type	other_2_kwh_type	Energy source description of kilowatt-hours produced from sources other than diesel/fuel oil or hydroelectric power. Description listed here corresponds to kilowatt-hours listed under 'other_2_kwh_generated'. Variable name in AEA's PCE NAVISION system: Other 2 kWh Type.
Other 2 kWh Generated	other_2_kwh_generated	Total kilowatt-hours produced from fuels different than diesel/fuel oil or hydroelectric power. For example, natural gas or wind. Please refer to 'other_2_kwh_type' for the corresponding fuel type description. Variable name in AEA's PCE NAVISION system: Other 2 kWh Generated.
Purchased From	purchased_from	Utility or independent power producer (IPP) name from which kilowatt-hours were purchased. Variable name in AEA's PCE NAVISION system: Purchased From.
Total kWh Purchased	kwh_purchased	Total kilowatt-hours purchased from a utility or independent power producer (IPP) and subsequently sold by the purchasing utility to its customers. Variable name in AEA's PCE NAVISION system: Total kWh Purchased.

Powerhouse Consumption kWh	powerhouse_consumption_kwh	Total monthly kilowatt-hours used by the powerhouse for its operations as reported by the utility to the Alaska Energy Authority. Also known as Station Service. Variable name in AEA's PCE NAVISION system: Powerhouse Consumption kWh.
Peak Consumption kW	peak_consumption_kw	Highest measured power demand in kilowatts for the month as reported by the utility to the Alaska Energy Authority. Variable name in AEA's PCE NAVISION system: Peak Consumption kW.
Residential Sold To	residential_kwh_sold	Total kilowatt-hours sold to all residential customer accounts reported by the utility to the Alaska Energy Authority. Variable name in AEA's PCE NAVISION system: Residential Sold To.
Commercial Sold To	commercial_kwh_sold	Total kilowatt-hours sold to all commercial customer accounts reported by the utility to the Alaska Energy Authority. Variable name in AEA's PCE NAVISION system: Commercial Sold To.
Com. Facil. Sold To	community_kwh_sold	Total kilowatt-hours sold to all community facilities customer accounts reported by the utility to the Alaska Energy Authority. Variable name in AEA's PCE NAVISION system: Com. Facil. Sold To.
Govt. Facil. Sold To	government_kwh_sold	Total kilowatt-hours sold to all government facilities customer accounts reported by the utility to the Alaska Energy Authority. Variable name in AEA's PCE NAVISION system: Govt. Facil. Sold To.
Unbilled Sold To	unbilled_kwh	Total kilowatt-hours consumed by customers but not billed for, as reported by the utility to the Alaska Energy Authority. Variable name in AEA's PCE NAVISION system: Unbilled Sold To.
Residential Customers	residential_customers	Total number of residential customer accounts billed by the utility as reported to the Alaska Energy Authority. Variable name in AEA's PCE NAVISION system: Residential Customers.
Commercial Customers	commercial_customers	Total number of commercial customer accounts billed by the utility as reported to the Alaska Energy Authority. Variable name in AEA's PCE NAVISION system: Commercial Customers.

Com. Facil. Customers	community_customers	Total number of community facilities customer accounts billed by the utility as reported to the Alaska Energy Authority. Variable name in AEA's PCE NAVISION system: Com. Facil. Customers.
Govt. Facil. Customers	government_customers	Total number of government customer accounts billed by the utility as reported to the Alaska Energy Authority. Variable name in AEA's PCE NAVISION system: Govt. Facil. Customers.
Unbilled Customers	unbilled_customers	Total number of customers that received electricity but were not billed by the utility as reported to the Alaska Energy Authority. Variable name in AEA's PCE NAVISION system: Unbilled Customers.
Other Customers	other_customers	Total number of customer accounts not pre-identified billed by the utility as reported to the Alaska Energy Authority. Variable name in AEA's PCE NAVISION system: Other Customers.
Other Customers Description	other_customers_description	Detailed description of customer type under 'other' category billed by the utility as reported to the Alaska Energy Authority. Variable name in AEA's PCE NAVISION system: Other Customers Description.
notes		Variable added by ISER. Information regarding one or multiple data points in the observation the user should know but not included in the variables listed.
Stage Code		Variable not included.
Vendor Invoice No.		Variable not included.
Document No.		Variable not included.
Ar Code		Variable not included.
Rate ID		Variable not included.
Check		Variable not included.
Dimension 6 Code		Variable not included.
Last Modified		Variable not included.
Last Modified By		Variable not included.
Date Created		Variable not included.
Document Date		Variable not included.
Posting Date		Variable not included.
Created By		Variable not included.